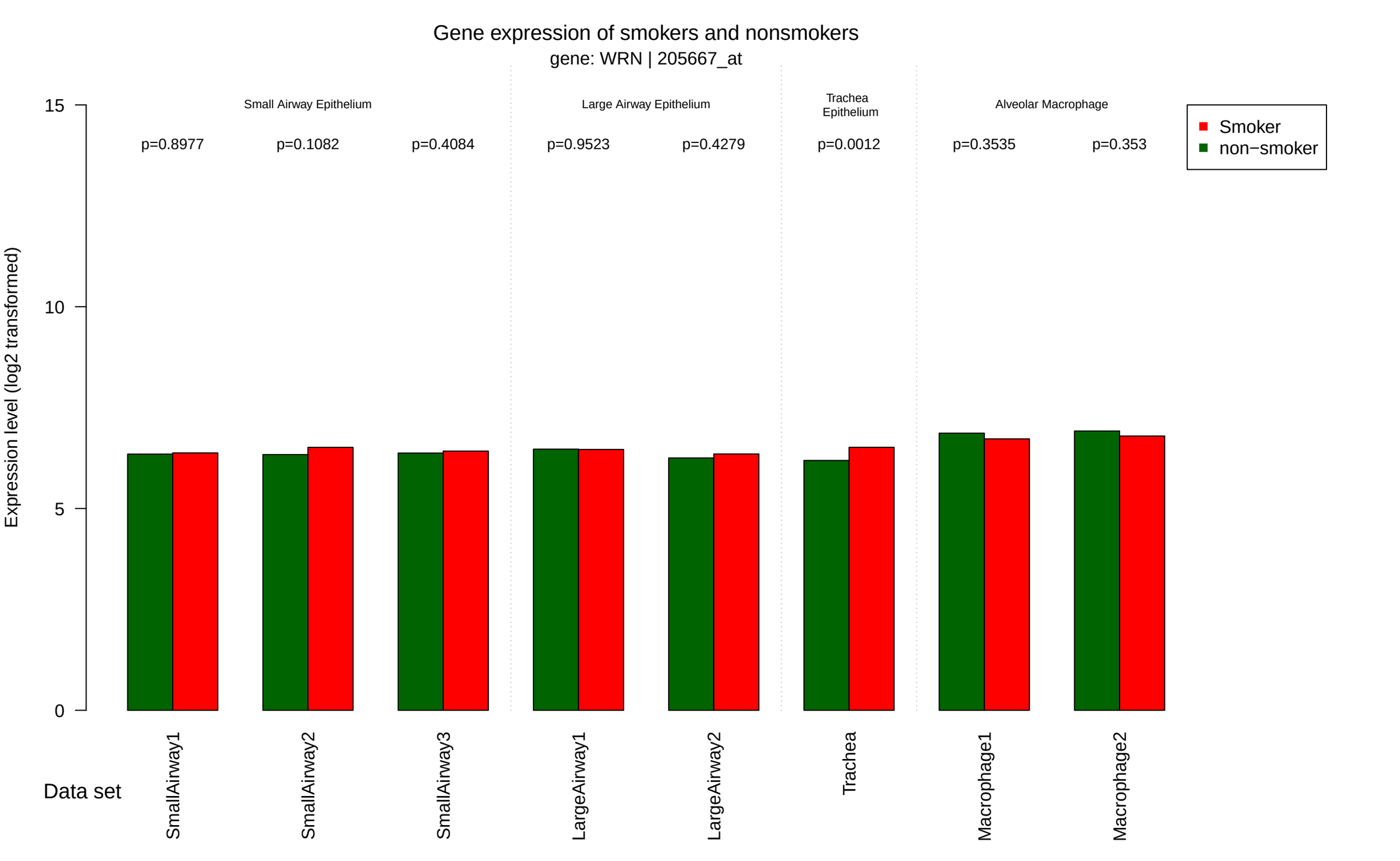
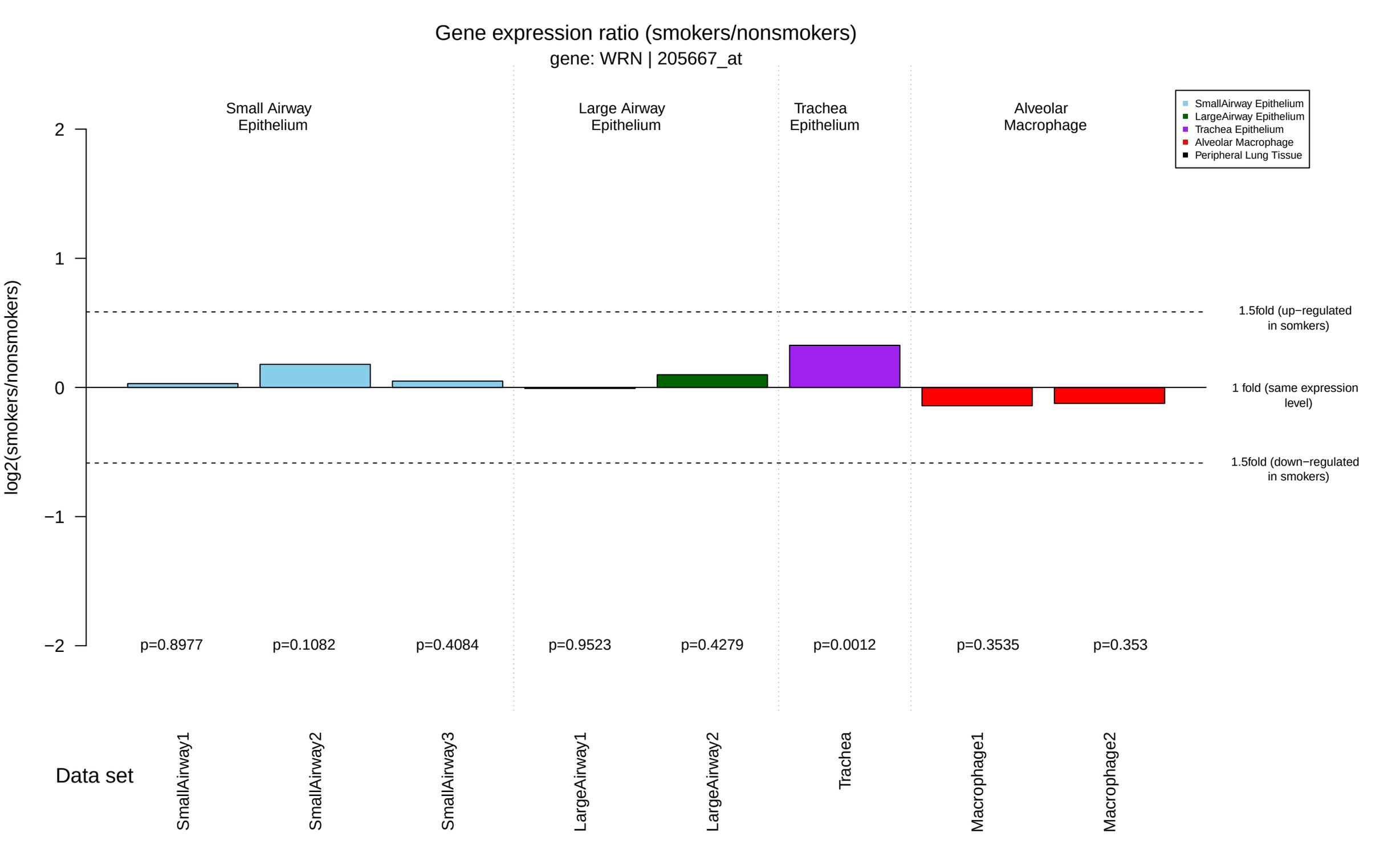
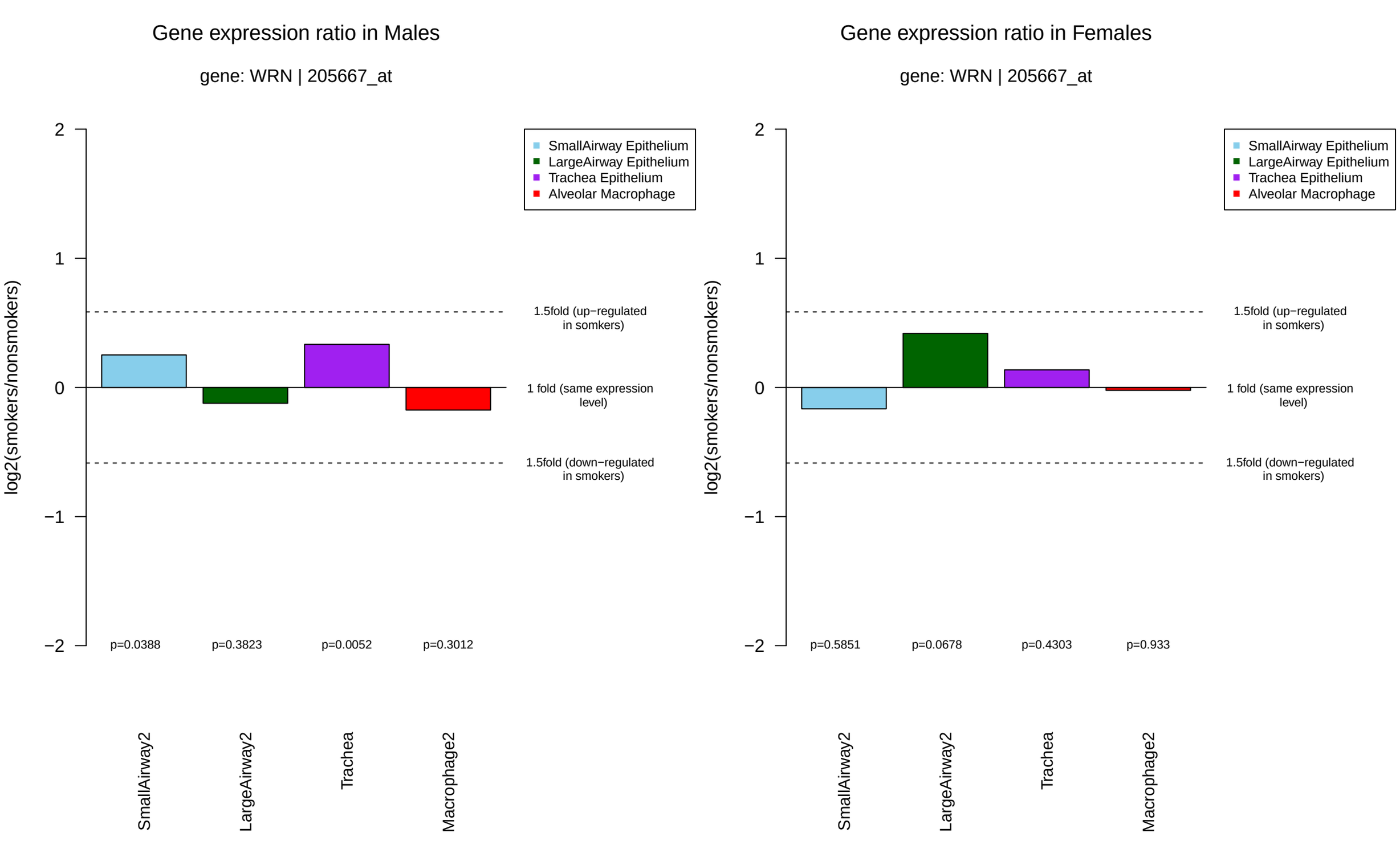
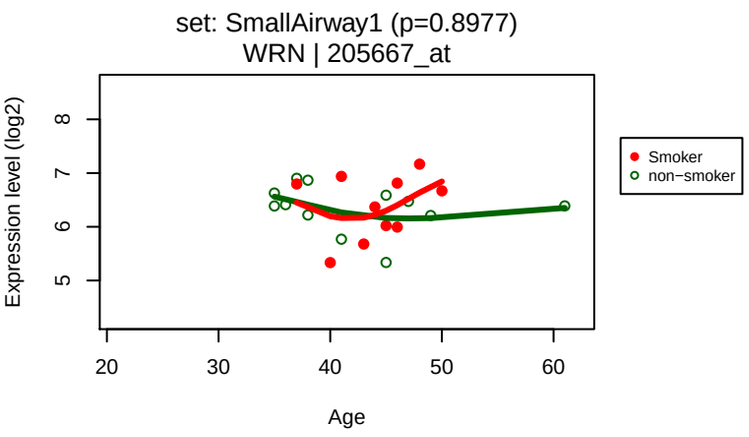
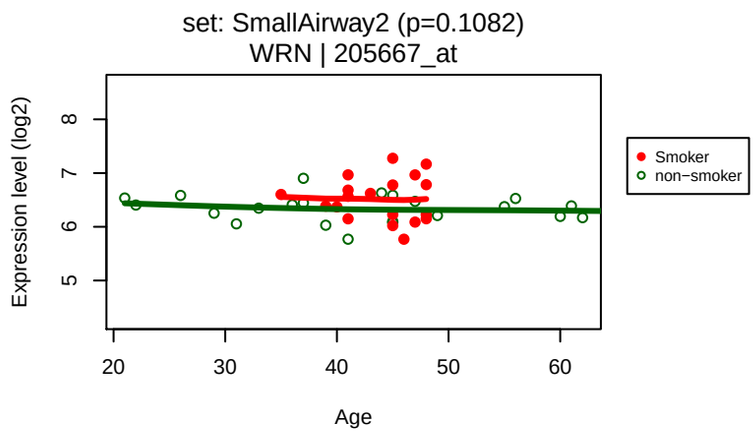
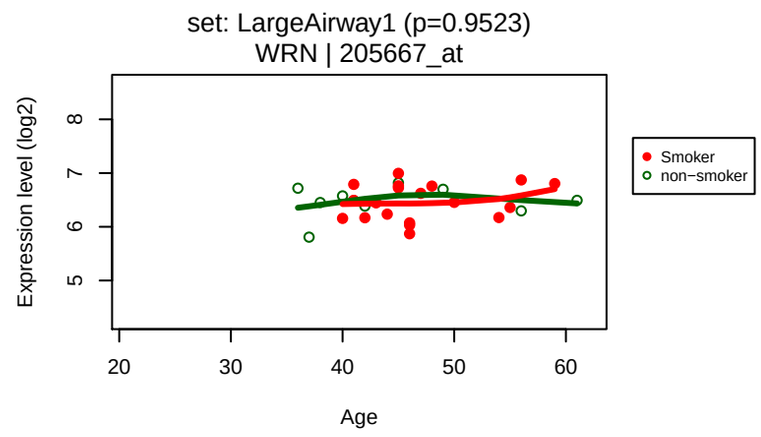

Gene expression of smokers and nonsmokers
gene: WRN | 205667_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8977
p=0.1082
p=0.4084
p=0.9523
p=0.4279
p=0.0012
p=0.3535
p=0.353
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

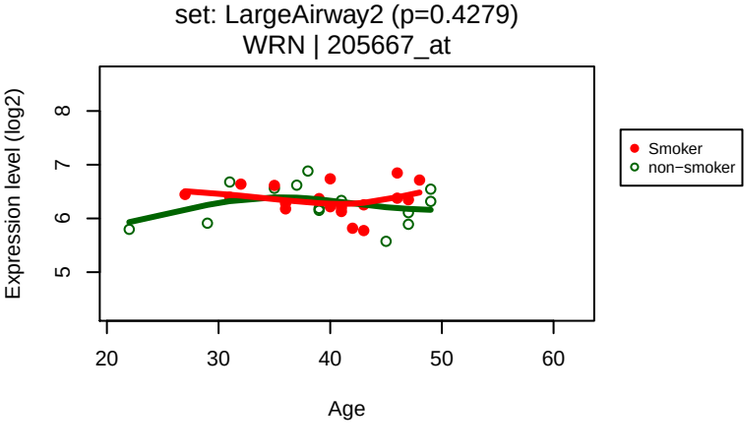

Gene expression ratio (smokers/nonsmokers)
gene: WRN | 205667_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8977
p=0.1082
p=0.4084
p=0.9523
p=0.4279
p=0.0012
p=0.3535
p=0.353
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

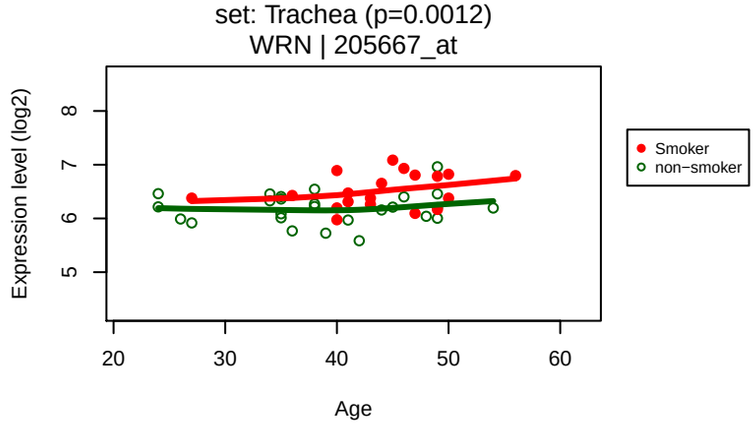

Gene expression ratio in Males
gene: WRN | 205667_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0388
p=0.3823
p=0.0052
p=0.3012
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WRN | 205667_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5851
p=0.0678
p=0.4303
p=0.933
SmallAirway2
LargeAirway2
Trachea
Macrophage2

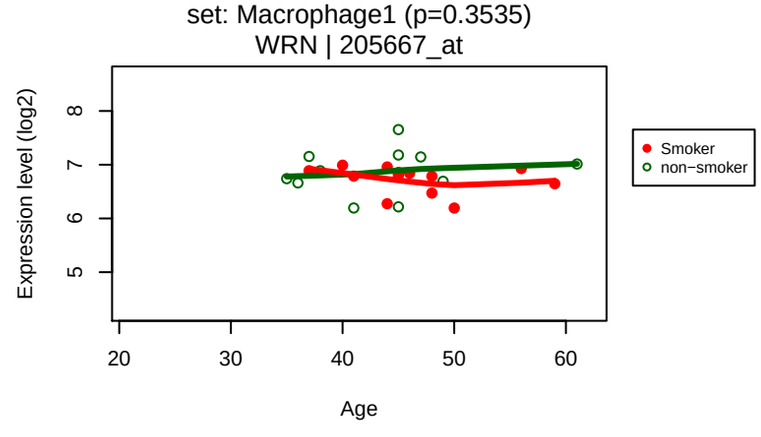

set: SmallAirway1 (p=0.8977)
WRN | 205667_at
Expression level (log2)
Age
Smoker
non−smoker

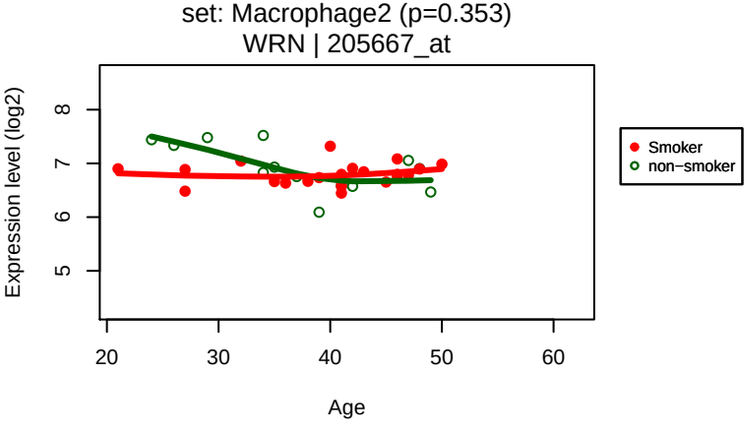

set: SmallAirway2 (p=0.1082)
WRN | 205667_at
Expression level (log2)
Age
Smoker
non−smoker

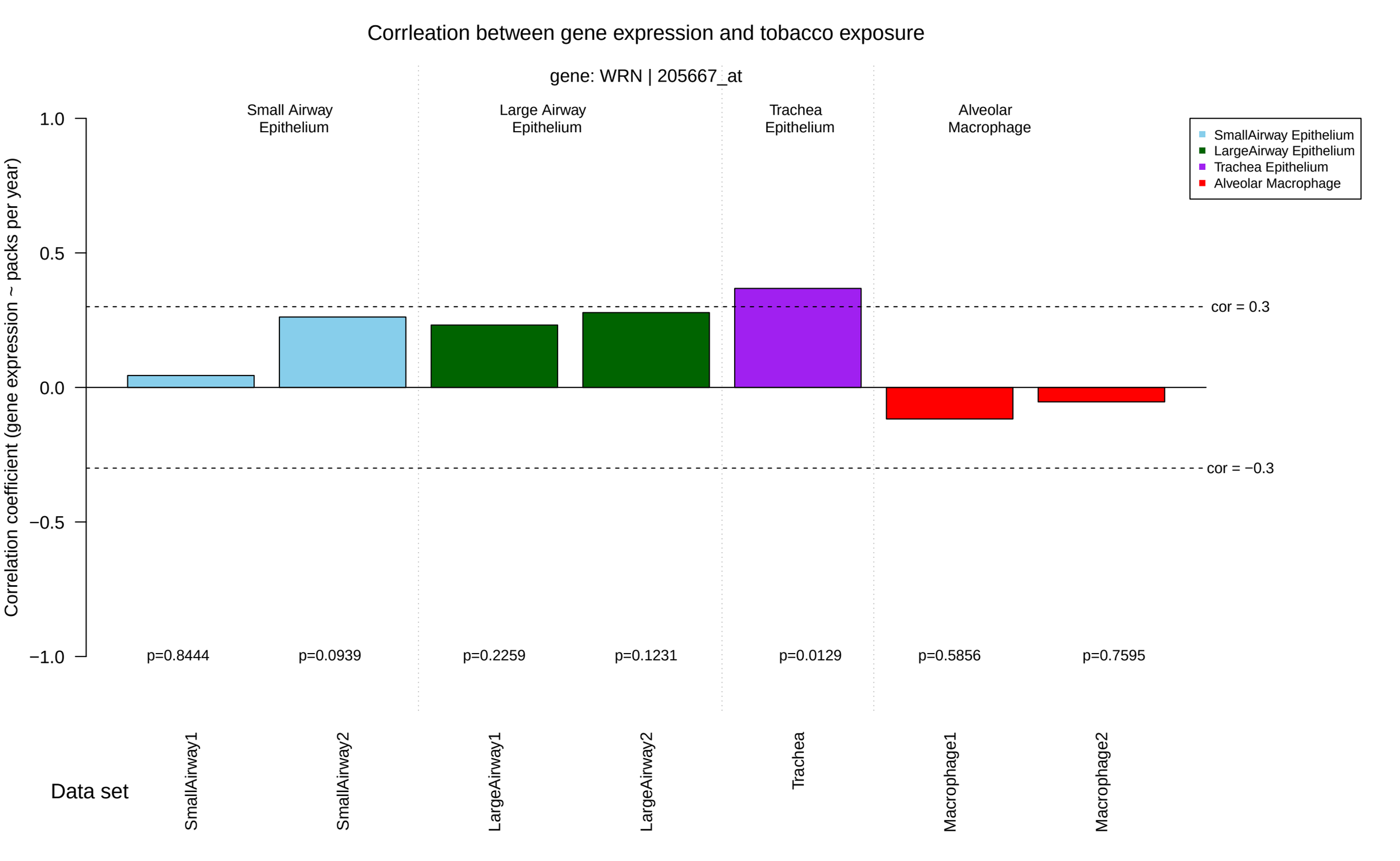

set: LargeAirway1 (p=0.9523)
WRN | 205667_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4279)
WRN | 205667_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0012)
WRN | 205667_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3535)
WRN | 205667_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.353)
WRN | 205667_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WRN | 205667_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8444
p=0.0939
p=0.2259
p=0.1231
p=0.0129
p=0.5856
p=0.7595
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2